Category Theory for Mathematical Physics — Exercise Sheet 4

Duality: cups, caps and quantum dimension

Variant: Solutions

Conventions as in Lecture 4: $\mathrm{ev}_X : X^* \otimes X \to 1$ is a cap and $\mathrm{coev}_X : 1 \to X \otimes X^*$ a cup (Definition 4.1); a pivotal structure a gives the traces of Definition 4.6; in $\mathrm{Vect}_{\mathrm{fd}}$, a is the canonical $V \to V^{**}$. In Exercise 2, for a qubit, $|0\rangle, |1\rangle$ is the standard basis, X and Z are the Pauli matrices (elsewhere X is an object), and $|\Phi^{\pm}\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$, $|\Psi^{\pm}\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$. The letter j is a gauge number in Exercise 4 and an isomorphism $V \to V^*$ in Exercises 5 and 6, as in Definition 4.9; spins are written ℓ, so the lecture's $(-1)^{2j}a$ is $(-1)^{2\ell}a$ here.

Exercise 1 (Rotated boxes, a second dual and the cyclic trace). (i) In $\mathrm{Vect}_{\mathrm{fd}}$ let $f : X \to Y$, with bases (x_k) of X and (y_l) of Y, dual bases (x^k), (y^l), and $f(x_k) = \sum_l f_{lk}\, y_l$. Show that the rotated box f^* of Equation (4.2) has the transpose matrix of f.

(ii) Let $X = \mathbb{C}^2$ and $G = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Take $X' = \mathbb{C}^2$ (column vectors) with $\mathrm{ev}'(u \otimes x) = u^T G x$. Find coev′ making $(X', \mathrm{ev}', \mathrm{coev}')$ a left dual of X (check both snakes), and compute the isomorphism $\varphi : X^* \to X'$ of Theorem 4.2(a), with X^* written as row vectors.

(iii) Let $f = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix} : \mathbb{C}^3 \to \mathbb{C}^2$ and $g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} : \mathbb{C}^2 \to \mathbb{C}^3$. Compute $\mathrm{tr}(g \circ f)$ and $\mathrm{tr}(f \circ g)$, and draw the three pictures of Equation (4.8) for this pair.

(iv) Cyclicity moves a box around the loop. Show that it does not allow other reorderings: find 2×2 matrices A, B, C with $\mathrm{tr}(ABC) \neq \mathrm{tr}(ACB)$. What, in the picture, stops C from jumping over B?

Solution. (i) The picture of f^* applied to y^l is $(\mathrm{ev}_Y \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{Y^*} \otimes f \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{Y^*} \otimes \mathrm{coev}_X)$: the cup gives $y^l \otimes \sum_k x_k \otimes x^k$, then f gives $y^l \otimes \sum_k f(x_k) \otimes x^k$, and the cap leaves $\sum_k y^l(f(x_k))\, x^k = \sum_k f_{lk}\, x^k$, where f_{lk} is the matrix of f. So the matrix of f^* in the dual bases has entry (k, l) equal to f_{lk}: the transpose, Lecture 1's dual map.

(ii) Write $\mathrm{coev}'(1) = \sum_i e_i \otimes c_i$ with $c_i \in X'$. The first snake, $(\mathrm{id}_X \otimes \mathrm{ev}') \circ (\mathrm{coev}' \otimes \mathrm{id}_X) = \mathrm{id}_X$, sends x to $\sum_i e_i\, c_i^T G x$, so it holds exactly when $\sum_i e_i c_i^T = G^{-1}$, that is $c_i = G^{-T} e_i$: $c_1 = e_1 - e_2$, $c_2 = e_2$. The second snake, $(\mathrm{ev}' \otimes \mathrm{id}_{X'}) \circ (\mathrm{id}_{X'} \otimes \mathrm{coev}')$, sends u to $\sum_i (u^T G e_i)\, G^{-T} e_i = G^{-T} G^T u = u$. By Theorem 4.2(a), $\varphi(\xi) = (\mathrm{ev} \otimes \mathrm{id}_{X'})(\xi \otimes \mathrm{coev}'(1)) = \sum_i \xi(e_i)\, G^{-T} e_i = G^{-T} \xi^T$, and indeed $\mathrm{ev}'(\varphi(\xi) \otimes x) = \xi G^{-1} G x = \xi(x)$. Any other dual of X is matched with X^* in the same way, by the one isomorphism that carries ev to ev′.

(iii) $fg = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $gf = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 1 & 3 & 3 \end{pmatrix}$, so $\mathrm{tr}(fg) = 5 = \mathrm{tr}(gf)$, although one is a 2×2 and the other a 3×3 matrix. The three pictures are Equation (4.8) with $X = \mathbb{C}^3$, $Y = \mathbb{C}^2$: f then g on the up-going strand; f on the up-going and $g^* = g^T$ on the down-going strand; g then f on the up-going strand (now on Y).

(iv) With $A = E_{12}$, $B = E_{21}$, $C = E_{11}$: $AB = E_{11}$ and $ABC = E_{11}$, of trace 1; but $AC = E_{12}E_{11} = 0$, so $\mathrm{tr}(ACB) = 0$. Cyclic rotations are allowed, $\mathrm{tr}(CAB) = \mathrm{tr}(E_{11}E_{12}E_{21}) = 1$. In the picture the three boxes sit on one strand closed into a loop; a box can travel around

the loop, through the cup and the cap, but two boxes on the same strand cannot pass each other (that would be composing in another order).

Exercise 2 (Bent wires and teleportation's four outcomes). (i) Draw the two maps of Equation (4.4), $\mathrm{Hom}(U \otimes V, W) \to \mathrm{Hom}(U, W \otimes V^*)$ and back, and show with one snake equation that bending a strand up and then down returns the morphism you started with. In $\mathrm{Vect}_{\mathrm{fd}}$ with $U = \mathbb{C}$, what does bending turn $f\colon \mathbb{C}^3 \to \mathbb{C}^2$ of Exercise 1 into?

Alice holds $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ and the first half of $|\Phi^+\rangle$; Bob holds the second half.

(ii) For each Bell state $|B\rangle \in \{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$ compute $(\langle B| \otimes \mathrm{id})(|\psi\rangle \otimes |\Phi^+\rangle)$, and write it as $\frac{1}{2}\sigma_B\,|\psi\rangle$.

(iii) Each Bell state is $(\mathrm{id} \otimes M_B)\,|\Phi^+\rangle$ for a 2×2 matrix M_B. Draw $\langle B|$ as the cap of Equation (4.5) with a box on its right leg, and explain σ_B by sliding that box along the snake to Bob's strand.

Solution. (i) The two maps are the pictures of Equation (4.4): $f \mapsto (f \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_U \otimes \mathrm{coev}_V)$ bends the V strand of f up to the right, and $g \mapsto (\mathrm{id}_W \otimes \mathrm{ev}_V) \circ (g \otimes \mathrm{id}_V)$ bends it back down. Bending f up and then down gives

$$(\mathrm{id}_W \otimes \mathrm{ev}_V) \circ (f \otimes \mathrm{id}_{V^*} \otimes \mathrm{id}_V) \circ (\mathrm{id}_U \otimes \mathrm{coev}_V \otimes \mathrm{id}_V),$$

and f slides up past the cap (interchange, Theorem 3.4): this is $f \circ (\mathrm{id}_U \otimes [(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V)]) = f$ by the snake equation for V; in pictures, the V strand is a snake that straightens. The other composite is the same picture turned over, and uses the other snake. With $U = \mathbb{C}$, $f\colon \mathbb{C}^3 \to \mathbb{C}^2$ becomes $(f \otimes \mathrm{id})(\mathrm{coev}(1)) = \sum_k f(x_k) \otimes x^k = \sum_{l,k} f_{lk}\, y_l \otimes x^k$: the same numbers f_{lk}, read as a vector in $\mathbb{C}^2 \otimes (\mathbb{C}^3)^*$.

(ii) Writing Alice's qubits first,

$$\begin{aligned}|\psi\rangle \otimes \left|\Phi^+\right\rangle = \tfrac{1}{2}\big[&\left|\Phi^+\right\rangle(\alpha\,|0\rangle + \beta\,|1\rangle) + |\Phi^-\rangle(\alpha\,|0\rangle - \beta\,|1\rangle)\\ &+ \left|\Psi^+\right\rangle(\alpha\,|1\rangle + \beta\,|0\rangle) + |\Psi^-\rangle(\alpha\,|1\rangle - \beta\,|0\rangle)\big],\end{aligned}$$

as one checks by expanding both sides on $|000\rangle, \ldots, |111\rangle$. Projecting with $\langle B| \otimes \mathrm{id}$ leaves Bob's factor: $\frac{1}{2}\,|\psi\rangle$, $\frac{1}{2}Z\,|\psi\rangle$, $\frac{1}{2}X\,|\psi\rangle$ and $\frac{1}{2}(\alpha\,|1\rangle - \beta\,|0\rangle) = \frac{1}{2}XZ\,|\psi\rangle$. Bob undoes them with I, Z, X and ZX.

(iii) $(\mathrm{id} \otimes M)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\sum_i |i\rangle \otimes M\,|i\rangle$ gives $|\Phi^-\rangle$ for $M = Z$, $|\Psi^+\rangle$ for $M = X$ and $|\Psi^-\rangle$ for $M = XZ$ ($XZ\,|0\rangle = |1\rangle$, $XZ\,|1\rangle = -|0\rangle$). So $\langle B| = \langle\Phi^+|\,(\mathrm{id} \otimes M_B^\dagger)$: Alice's cap with the box $M_B^\dagger$ on its right leg, the leg that meets the cup. That box slides along the snake: through the cup it becomes the transpose on Bob's leg, since $(M \otimes \mathrm{id})\,|\Phi^+\rangle = (\mathrm{id} \otimes M^T)\,|\Phi^+\rangle$ (the rotated box of Exercise 1(i), in the orthonormal basis). The snake then straightens as in Equation (4.5), leaving $\frac{1}{2}(M_B^\dagger)^T\,|\psi\rangle = \frac{1}{2}\overline{M_B}\,|\psi\rangle$ on Bob's strand. The four M_B are real, so $\sigma_B = M_B$: I, Z, X, XZ, as in (ii).

Exercise 3 (Super vector spaces). The objects of sVec are the finite-dimensional $\mathbb{Z}_2$-graded spaces $V = V_0 \oplus V_1$; its morphisms are the linear maps preserving the grading. It has the tensor product with $(V \otimes W)_0 = V_0 \otimes W_0 \oplus V_1 \otimes W_1$, $(V \otimes W)_1 = V_0 \otimes W_1 \oplus V_1 \otimes W_0$, the unit $\mathbb{C}$ in degree 0, and the duals, ev and coev of $\mathrm{Vect}_{\mathrm{fd}}$, with $(V^*)_i = (V_i)^*$. Let P_V be $+1$ on V_0 and -1 on V_1.

(i) Show that $a'_V = a_V \circ P_V$ is a pivotal structure.

(ii) Show that its left trace is the *supertrace*, $\mathrm{str}(h) = \mathrm{tr}(h|_{V_0}) - \mathrm{tr}(h|_{V_1})$, so that $d_V = \dim V_0 - \dim V_1$, and that its right trace is also the supertrace.

(iii) Draw the picture proof of $d_{V\otimes W} = d_V d_W$ (Theorem 4.7): the loop on $V \otimes W$ as two nested loops. Then check $d_{V\otimes W} = d_V d_W$ directly from the grading of $V \otimes W$, and find a nonzero object of dimension 0.

Lecture 7 adds the braiding that makes this category "super"; Lecture 10 uses the pivotal structure a on it instead, with $d = +1$ on the odd line.

Solution. (i) P_V is an isomorphism, so a'_V is. A morphism $f : V \to W$ maps V_i into W_i, so $f \circ P_V = P_W \circ f$, and with the naturality of a, $f^{**} \circ a'_V = f^{**} \circ a_V \circ P_V = a_W \circ f \circ P_V = a_W \circ P_W \circ f = a'_W \circ f$. On $V \otimes W$ the degree of $v \otimes w$ is the sum of the degrees, so $P_{V\otimes W} = P_V \otimes P_W$, and $a_{V\otimes W} = a_V \otimes a_W$ gives $a'_{V\otimes W} = a'_V \otimes a'_W$.

(ii) Take a basis (e_i) of V in which each e_i is homogeneous. The left trace is $\widetilde{\mathrm{ev}}_V \circ (h \otimes \mathrm{id}) \circ \mathrm{coev}_V$ with $\widetilde{\mathrm{ev}}_V(v \otimes \varphi) = \mathrm{ev}_{V^*}(a'_V(v) \otimes \varphi) = \varphi(P_V v)$. So $\mathrm{tr}_L(h) = \sum_i e^i(P_V h e_i) = \mathrm{tr}(P_V h)$, and since h preserves the grading this is $\mathrm{tr}(h|_{V_0}) - \mathrm{tr}(h|_{V_1})$. With $h = \mathrm{id}$, $d_V = \dim V_0 - \dim V_1$. The right trace uses $\widetilde{\mathrm{coev}}_V = (\mathrm{id} \otimes a'^{-1}_V) \circ \mathrm{coev}_{V^*}$, which sends 1 to $\sum_i e^i \otimes P_V e_i$ ($P^{-1} = P$); then $\mathrm{tr}_R(h) = \sum_i e^i(h P_V e_i) = \mathrm{tr}(h P_V) = \mathrm{str}(h)$ as well. So the structure is spherical. (The lecture's $a_s = -\mathrm{id}$ on $\mathrm{Vec}_{\mathbb{Z}_2}$ is the same choice, on the one-dimensional odd object; Lecture 10 uses a itself, with $d = +1$ there. Both are pivotal structures; this one gives the super-dimension.)

(iii) Draw the strand of $V \otimes W$ as two parallel strands, V on the left. The cup and the cap of $V \otimes W$ are the nested ones (the cup of W inside the cup of V, and likewise the caps, with $a'_{V\otimes W} = a'_V \otimes a'_W$ by (i)), so the loop on $V \otimes W$ is a W loop inside a V loop. The inner loop is the scalar d_W, a number since $\otimes$ is bilinear, so it moves out of the picture (Equation (3.6) and the remark after it), leaving d_W times the V loop, $d_V d_W$. Directly: $\dim(V \otimes W)_0 - \dim(V \otimes W)_1 = v_0 w_0 + v_1 w_1 - v_0 w_1 - v_1 w_0 = (v_0 - v_1)(w_0 - w_1)$, with $v_i = \dim V_i$, $w_i = \dim W_i$, as Theorem 4.7 requires. The object $\mathbb{C}^{1|1}$, with $V_0 = V_1 = \mathbb{C}$, is nonzero and has $d = 1 - 1 = 0$.

Exercise 4 (The semion snake). In Vec^ω_G on $\mathbb{Z}_2 = \{1, s\}$ with $\omega(s,s,s) = -1$, $s^* = s$ and $s \otimes s = 1$, so a cup $\mathrm{coev}_s = \lambda\ \mathrm{id}_1$ and a cap $\mathrm{ev}_s = \mu\ \mathrm{id}_1$ are numbers.

(i) Show that the first snake of Equation (4.1) is $\lambda\mu\,\omega(s,s,s)\ \mathrm{id}_s$, so both snakes are straight exactly when $\lambda\mu = -1$. What is the snake for the naive choice $\lambda = \mu = 1$?

(ii) A *gauge* changes the identification $s \otimes s = 1$ by a number $j \neq 0$: it is the monoidal functor that is the identity on objects with $J_{s,s} = j$ and the other J's and J_0 equal to 1. It sends (λ, μ) to $(\lambda/j, \mu j)$. Show that it is a monoidal functor, and that no gauge makes $\lambda = \mu = 1$ a solution. Is there a solution with $\lambda = \mu$?

Solution. (i) The first snake is $(\mathrm{id}_s \otimes \mathrm{ev}_s) \circ \alpha_{s,s,s} \circ (\mathrm{coev}_s \otimes \mathrm{id}_s)$; on the simple object s every factor is a scalar, and they multiply to $\lambda\,\omega(s,s,s)\,\mu = -\lambda\mu$ (as in Example 4.4 with $\mathrm{coev}_s = \mathrm{id}$). The second snake passes through $\alpha^{-1}_{s,s,s}$ and gives $\mu\,\omega(s,s,s)^{-1}\lambda = -\lambda\mu$ as well. Both are id_s exactly when $\lambda\mu = -1$. The naive choice $\lambda = \mu = 1$ gives the snake $-\mathrm{id}_s$.

(ii) With $J_{s,s} = j$ and $J_{1,g} = J_{g,1} = J_0 = 1$, the first condition of Definition 3.10 at (g,h,k) compares $\omega(g,h,k)$ times two J's on each side; at (s,s,s) it reads $\omega\, j \cdot 1 = 1 \cdot j\,\omega$, and at the other triples every J that appears is 1 or both sides have the same factors; the unit conditions hold since $J_0 = J_{1,g} = J_{g,1} = 1$. So it is a monoidal functor from Vec^ω_G to itself. It transports the cap to $\mathrm{ev}_s \circ J_{s,s}$ and the cup to $J^{-1}_{s,s} \circ \mathrm{coev}_s$, that is $(\lambda, \mu) \mapsto (\lambda/j, \mu j)$, and the product $\lambda\mu$ does not change. So $\lambda\mu = -1$ in every gauge, and $\lambda = \mu = 1$ is never a solution: the sign $-1 = \omega(s,s,s)$ is not a choice of normalisation (compare Proposition 3.11). A solution with $\lambda = \mu$ needs $\lambda^2 = -1$: $\lambda = \mu = \pm i$.

Exercise 5 (Frobenius–Schur indicators of Q_8 and D_8). Let $Q_8 \subset \mathrm{SU}(2)$ be generated by

$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and D_8, the symmetries of a square, by the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and the reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Each acts on $V = \mathbb{C}^2$ by these matrices, irreducibly.

(i) Show that both groups have order 8, and that the two characters on V take the value 2 once, -2 once and 0 six times.

(ii) Compute $\kappa_V = \frac{1}{8}\sum_g \chi_V(g^2)$ for each (the character formula of *Symmetries*).

(iii) Find a nonzero invariant bilinear form on V for each group, and check against Definition 4.9 that a nonzero symmetric invariant form gives $\kappa = +1$ and a nonzero antisymmetric one $\kappa = -1$. The two characters take the same values, with the same multiplicities; what group data, beyond these values, does κ see?

Solution. (i) With $q_1 = \mathrm{diag}(i, -i)$ and $q_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $q_1^2 = q_2^2 = -1$ and $q_1 q_2 = -q_2 q_1 =: q_3$, so $Q_8 = \{\pm 1, \pm q_1, \pm q_2, \pm q_3\}$, of order 8; the traces are 2 at 1, -2 at -1 and 0 at the six others. The rotation R by $\pi/2$ and the reflection S generate the four rotations $1, R, R^2 = -1, R^3$ and the four reflections S, SR, SR^2, SR^3; the traces are 2, 0, -2, 0 for the rotations and 0 for every reflection. So both characters take the values $2, -2, 0$ with multiplicities $1, 1, 6$ (on matching classes: the identity, the central -1, and three classes of size two).

(ii) In Q_8, $g^2 = 1$ only for $g = \pm 1$, and $g^2 = -1$ for the other six: $\sum_g \chi(g^2) = 2\cdot 2 + 6\cdot(-2) = -8$, so $\kappa = -1$. In D_8, $g^2 = 1$ for 1, R^2 and the four reflections, and $g^2 = R^2 = -1$ for R, R^3: $\sum_g \chi(g^2) = 6\cdot 2 + 2\cdot(-2) = 8$, so $\kappa = +1$.

(iii) D_8 is made of real orthogonal matrices, so $B(v, w) = v^T w$ satisfies $B(gv, gw) = B(v, w)$: symmetric. Q_8 lies in SU(2), so the form $\Omega(v, w) = v_1 w_2 - v_2 w_1$ satisfies $\Omega(gv, gw) = \det(g)\,\Omega(v, w) = \Omega(v, w)$: antisymmetric. For a nonzero invariant form B, the map $j(v) = B(\cdot, v)$ is an intertwiner $V \to V^*$; its kernel is a subrepresentation, not all of V, so it is 0 (V is irreducible) and j is an isomorphism of representations; and $(j^* \circ a_V)(v) = a_V(v) \circ j = B(v, \cdot)$. So $j^* \circ a_V = j$ when B is symmetric and $= -j$ when B is antisymmetric: $\kappa = +1$ for D_8 and $\kappa = -1$ for Q_8, as in (ii). The two-dimensional representations of Q_8 and D_8 have the same character values, with the same multiplicities, but different indicators. A character as a function on its group does determine κ, by the formula of (ii); what the values alone miss is the squaring map $g \mapsto g^2$, which differs (2 elements square to 1 in Q_8, 6 in D_8). The indicator sees whether the invariant form is symmetric or antisymmetric.

Exercise 6 ⋆. *The Penrose cap and the pivotal structure.* Let $V = \mathbb{C}^2$ be spin-$\frac{1}{2}$, with $\Omega(e_1, e_2) = 1 = -\Omega(e_2, e_1)$ as in Theorem 4.8, and $j\colon V \to V^*$, $j(v) = \Omega(\cdot, v)$.

(i) Show that $(\mathrm{id} \otimes j^{-1}) \circ \mathrm{coev}_V$ is the cup Ω and that $\mathrm{ev}_V \circ (j \otimes \mathrm{id})$ is the cap $-\Omega^\dagger$: the straight pair of Theorem 4.8 is (coev, ev) moved along j.

(ii) Show that the loop $-\Omega^\dagger \circ \Omega = -2$ is the quantum dimension $\mathrm{tr}_L(\mathrm{id}_V)$ for the map $a''_V\colon V \to V^{**}$ with $a''_V(v)(\varphi) = \Omega(j^{-1}\varphi, v)$, and that $a''_V = -a_V$.

(iii) On the finite-dimensional representations of SU(2) let $a'_W = a_W \circ \rho_W(-1)$. Show that a' is a pivotal structure, equal to a'' on spin-$\frac{1}{2}$; compute d_W for a' on spin ℓ, and the Frobenius–Schur indicator of spin-$\frac{1}{2}$ for a'.

Solution. (i) $j(e_1) = \Omega(\cdot, e_1)$ takes $e_1 \mapsto 0$, $e_2 \mapsto -1$, so $j(e_1) = -e^2$; and $j(e_2) = e^1$. Hence $j^{-1}(e^1) = e_2$, $j^{-1}(e^2) = -e_1$, and $(\mathrm{id} \otimes j^{-1})(e_1 \otimes e^1 + e_2 \otimes e^2) = e_1 \otimes e_2 - e_2 \otimes e_1 = \Omega$. On the other side, $\mathrm{ev}_V(j(v) \otimes w) = j(v)(w) = \Omega(w, v) = -\Omega(v, w)$, and $\Omega^\dagger(e_a \otimes e_b) = \Omega(e_a, e_b)$ for these real matrices, so $\mathrm{ev}_V \circ (j \otimes \mathrm{id}) = -\Omega^\dagger$. Moving a dual along an isomorphism keeps the snakes straight (insert $j^{-1}j$ in the middle of each snake), which is Theorem 4.8's statement that $(\Omega, -\Omega^\dagger)$ is straight.

(ii) The loop is $-\Omega^\dagger\circ\Omega = \mathrm{ev}_V\circ(j\otimes j^{-1})\circ\mathrm{coev}_V = \sum_i j(e_i)\big(j^{-1}(e^i)\big) = (-e^2)(e_2)+e^1(-e_1) = -2$. It is a cap on $V\otimes V^*$ closed against coev_V, that is $\widetilde{\mathrm{ev}}_V\circ\mathrm{coev}_V$ with $\widetilde{\mathrm{ev}}_V(v\otimes\varphi) = j(v)(j^{-1}\varphi)$. Writing $\widetilde{\mathrm{ev}}_V = \mathrm{ev}_{V^*}\circ(a''_V\otimes\mathrm{id})$ means $a''_V(v)(\varphi) = j(v)(j^{-1}\varphi) = \Omega(j^{-1}\varphi, v)$, so the loop is $\mathrm{tr}_L(\mathrm{id}_V)$ for a''. For $\varphi = j(w)$: $a''_V(v)(j(w)) = \Omega(w,v) = -\Omega(v,w) = -j(w)(v) = -a_V(v)(j(w))$. Since j is onto V^*, $a''_V = -a_V$. (Equivalently $j^*\circ a_V = -j$: $\kappa_{1/2} = -1$ for a, as in Lecture 4.)

(iii) -1 is central in SU(2), so each $\rho_W(-1)$ is an intertwiner, natural in W: $T\rho_W(-1) = \rho_{W'}(-1)T$ for every intertwiner T. Hence a' is natural. On $W\otimes W'$, $\rho(-1)$ acts as $\rho_W(-1)\otimes\rho_{W'}(-1)$, so $a'_{W\otimes W'} = a'_W\otimes a'_{W'}$: a' is a pivotal structure. On spin ℓ, $\rho(-1) = (-1)^{2\ell}$ (the irreducible representation of dimension $2\ell+1$ has -1 acting by $(-1)^{2\ell}$), so on spin-$\frac{1}{2}$, $a' = -a = a''$. As in Exercise 3, $d_W = \mathrm{tr}_L(\mathrm{id}_W) = \mathrm{tr}\,\rho_W(-1) = (-1)^{2\ell}(2\ell+1)$: -2 on spin-$\frac{1}{2}$, the Penrose loop, and 3 on spin 1. The indicator for a' is $j^*\circ a'_V = j^*\circ a_V\circ\rho_V(-1) = (-j)\circ(-\mathrm{id}) = j$: $\kappa' = +1$. The straight Penrose cap and its loop -2 belong to the pivotal structure $(-1)^{2\ell}a$, for which spin-$\frac{1}{2}$ has indicator +1.